Certified optimization and weak-signal tensor inference

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Status: 2026-09-11. These are supplied derivations, not independent review or machine-verified proofs. They build on the all-budget rates in `norm_loading_extension/` and preserve the previous stages. Results below do not assert a polynomial minimax algorithm for diffuse weak signals.

Observe the entire tensor $Z = A + \tau G$, with iid standard Gaussian entries, known $\tau > 0$, and $A \in \Theta(M, \bar{r})$, a Tucker or TT Frobenius ball. The functional is $\psi_L(A) = \langle L, A\rangle$; normalization of L is unnecessary. Write $J = J(\bar{r})$ for the existing canonical expansion bound: $A = \sum_{j\le J} a_j U_j$, each U_j a unit product tensor, and $\sum_j a_j^2 = \|A\|_F^2$. Zero terms can be appended. Assume positive caps.

For reference, the previously established equal/flat-loading rate is $\mathfrak{r}_K(m, M, \tau) = \tau\sqrt{\ell_m/m}\, g_K(x)$, where $\ell_m = \log(em)$, $x = M/(\tau\sqrt{\ell_m})$, and

$$g_K(x) = \begin{cases} \min\{x, 1 + \max(0, \log x), \ell_m\}, & K = 3, \\ \min\{x, 1\}, & K \ge 4. \end{cases}$$

The original equal diagonal has $m = p$ and weights $1/\sqrt{p}$. Length criteria below mean the infimum expected length at zero ($\mathcal{L}_0$), or the infimum worst-case expected length ($\mathcal{L}_{\max}$), over intervals honest on the stated parameter class.

C1. A quantitative contract for approximate minimax optimization

Use every support and every direction in the calibrated finite product nets. For their unit directions U and thresholds $\lambda_{s(U)}$, put

$$F_Z(B) = \max\{0, \max_U(|\langle U, Z - B\rangle| - \lambda_{s(U)})\}.$$

This function is 1-Lipschitz in Frobenius norm. Suppose an algorithm returns $\widehat{B} \in \Theta$ and numbers $l \le \inf_{B\in\Theta} F_Z(B) \le F_Z(\widehat{B}) = u$, with $u - l \le \varepsilon$. On the simultaneous noise event, the true A is feasible, so $F_Z(A) = 0$ and $u \le \varepsilon$. Consequently, for every support size s,

$$N_S(\widehat{B} - A) \le 2\lambda_s + \varepsilon.$$

With $J_2 = J(2\bar{r})$, the preceding ordered-diagonal proof gives

$$|\widehat{B}_{i,\dots,i} - A_{i,\dots,i}|_{(s)} \le \min\{2J_2^{(K-1)/2}M, 4J_2^{K/2}(\lambda_s + \varepsilon/2)\}s^{-K/2}.$$

For known orthogonal loading weights w_i, sorted by absolute value, the sum of the right side against w_s^* is an honest radius about $\psi_L(\widehat{B})$. It may be intersected with the structural range. Choosing $\varepsilon \le c\tau$ retains the existing all-budget minimax orders for equal/flat loadings. Indeed every λ_s is bounded below by a positive constant times τ, so the added term changes the existing minimum by only a fixed factor, including in the norm-limited regime.

For squared error set the scan error budget to 1/8. If the optimizer guarantees $\varepsilon \le c\tau$ for *every observation*, the same excess tail proof as before applies: $F_Z(\widehat{B}) \le F_Z(A) + \varepsilon$ and $N_S(\widehat{B} - A) \le 2\lambda_s + 2F_Z(A) + \varepsilon$. Thus the

existing all-budget MSE bound follows. A budget-limited run with a larger reported gap does not establish that MSE guarantee.

Endpoint version. Let $\mathcal{F} = \{B \in \Theta : F_Z(B) = 0\}$. If certified outer endpoints enclose $\psi_L(\mathcal{F})$, they define an honest interval, even at an arbitrary computation stop. If feasible witnesses establish each outer endpoint within ε of the exact endpoint, its length exceeds the ideal projection length by at most 2ε. An inner optimization solution alone does not give an honest interval. On a certified empty feasible set use $\{0\}$, as before.

Proof. Lipschitzness follows from $\|U\|_F = 1$, the reverse triangle inequality, and stability of finite maxima. The objective-gap implication uses $F_Z \geq 0$. Rank caps double under subtraction; the norm budget doubles as well. The ordered bound is exactly the earlier net comparison and diagonal quasi-norm inequality with $2\lambda_s$ replaced by $2\lambda_s + \varepsilon$. The Gaussian union event has probability at least $1 - \alpha$. Endpoint containment proves coverage on the same event. For the length assertion, use an inner feasible witness at each end. The squared-risk argument invokes the already supplied excess-tail integration, with only its deterministic tolerance constant changed. No algorithmic complexity conclusion follows from this contract.

C2. An implemented global solver on the binary rank-one class

For $p = 2$ and any fixed order $K \geq 3$, every rank-one tensor of norm at most M has a representation

$$B(a, \theta) = a \bigotimes_{k=1}^{K} (\cos\theta_k, \sin\theta_k), \quad a \in [-M, M], \quad \theta_k \in [0, \pi].$$

Each vector is represented up to sign on a semicircle; the product of signs is absorbed into a. The zero tensor is included. There is no grid restriction on the parameter space.

The code includes both singleton supports and the full support. For the latter it uses $n_K = \lceil \pi K \rceil$ equally spaced angles in $[0, \pi)$ per mode. Every unit vector is within Euclidean distance $1/(2K)$ of a net point up to sign: its angular discrepancy modulo π is at most $\pi/(2n_K)$. Absolute contractions remove the signs. The same telescoping proof therefore gives $\rho \leq 2N$. There are $D_1 = 2$ singleton directions and $D_2 = n_K^K$ full-support directions. Set

$$\lambda_s = \tau\sqrt{2\log(4D_s/\alpha)}, \qquad s = 1, 2.$$

The sum of the two Gaussian union bounds is at most α.

For each parameter box, exact real-arithmetic ranges of cosine and products enclose every scanned contraction and tensor entry. If their contraction range is $[b_U^-, b_U^+]$, then

$$l_{\text{box}} = \max\{0, \max_U [\text{dist}(\langle U, Z\rangle, [b_U^-, b_U^+]) - \lambda_{s(U)}]\}$$

is a lower bound for F_Z throughout the box. Centers provide actual rank-one upper witnesses. A largest-scaled-side split covers the parent with two children. Branch and bound retains global lower and upper bounds; the objective run ends at its requested gap or reports a budget stop. It implements the ideal residual criterion itself on this class.

For confidence projection reject a box only when its objective lower bound is positive. Loading-entry interval arithmetic bounds every surviving box's target; their hull contains the ideal confidence projection. Feasible amplitude intervals at fixed central factors supply inner witnesses. An endpoint gap is reported only when such witnesses exist. Endpoints need not reach a requested tolerance within a finite node budget, especially near singular feasibility boundaries. This is recorded, not treated as a successful exact optimization.

The code supports arbitrary loadings in the binary tensor, but caps above one and mode dimensions above two are not implemented. Exhaustive scans and global subdivision are exponential. Tests check range containment and global certificates against independent finite calculations. Floating point evaluations include conservative guards; these are numerical certificates under the derived inequalities, not formal directed-rounding interval proofs. The real-arithmetic method, rather than any particular floating point tolerance, is the mathematical result.

L1. Sharp small-budget rates for every loading tensor

Define the product overlap

$$\rho(L) = \sup_{\|u_k\|_2=1} |\langle L, \bigotimes_k u_k \rangle|.$$

For fixed positive caps, fixed $0 < \alpha < 1/4$, arbitrary mode dimensions, and *every* known loading tensor L, if $0 < M \le \tau$, then

$$\mathcal{L}_0(\Theta; L) \asymp \mathcal{L}_{\max}(\Theta; L) \asymp M\rho(L), \qquad \inf_{\widehat{\psi}} \sup_{\Theta} \mathbb{E}_A(\widehat{\psi} - \psi_L(A))^2 \asymp M^2\rho(L)^2.$$

Constants depend only on the fixed caps/order and confidence, not on L or the ambient dimensions. The zero loading has identically zero risk and length. More generally, for all $M, \tau > 0$,

$$c_\alpha \rho(L) \min(M, \tau) \le \mathcal{L}_0 \le \mathcal{L}_{\max} \le 2\min\{\sqrt{J}M\rho(L),\ z_{1-\alpha/2}\tau\|L\|_F\}.$$

Proof. The canonical expansion and Cauchy–Schwarz give $|\psi_L(A)| \le \rho(L)\sum_j |a_j| \le \sqrt{J}M\rho(L)$. This gives a deterministic interval and the risk of estimator zero. The direct Gaussian interval gives the other upper bound; choose between the two using known quantities. Compactness gives a maximizing product U; choose its sign so $\langle L, U\rangle = \rho(L)$. Both 0 and tU belong to every positive cap class, for $t = \min(M, \tau)/2$. Their Gaussian total variation is $2\Phi(t/(2\tau)) - 1 \le 2\Phi(1/4) - 1 < 1/4$. Honesty at both parameters implies $\mathbb{E}_0|C| \ge t\rho(L)(1 - 2\alpha - \mathrm{TV})$. The two-point squared-loss lower bound is a fixed positive multiple of $t^2\rho(L)^2$: classify an estimate by its nearest target. A classification error entails squared error at least $t^2\rho(L)^2/4$, while the sum of the two testing errors is at least $1 - \mathrm{TV}$. Therefore the larger squared risk is at least $t^2\rho(L)^2(1 - \mathrm{TV})/8$. For $M \le \tau$ the bounds match. This proof is a direct specialization of standard Gaussian two-point reasoning, not a novel general inference principle.

Limits. Computing $\rho(L)$ for a general high-order loading is itself a global optimization problem. A certified upper bound can replace it in the interval; a certified product witness supplies a lower bound. Without a constant ratio between them, an efficient sharp algorithm has not been obtained. This theorem covers all loading tensors in the stated budget regime; it does not characterize their larger-budget rates. The flat loading small-budget theorem extends to $M \lesssim \tau\sqrt{\log m}$, a larger regime than the arbitrary-loading result here.

L2. An optimized common-target lower bound for unequal weights

For an orthogonal product loading with decreasing nonzero absolute weights $w_1 \ge \cdots \ge w_m > 0$, choose $d \le m$ and $0 < \Delta \le Mw_d$. For signed original weights $\widetilde{w}_i$, set $A_i = (\Delta/\widetilde{w}_i)U_i$, $1 \le i \le d$. All alternatives have target Δ, satisfy the norm budget, and have rank one. They are orthogonal as full tensors. Let

$$v_i = \exp\{\Delta^2/(\tau^2 w_i^2)\} - 1, \quad H_d(\Delta) = \sum_{i=1}^{d} v_i^{-1}, \quad \pi_i = v_i^{-1}/H_d(\Delta).$$

The optimal chi-square divergence among mixtures on these alternatives is

$$\chi^2(Q, P_0) = \min_{\pi \in \mathcal{S}_d} \sum_i \pi_i^2 v_i = H_d(\Delta)^{-1}.$$

Thus every interval honest on the full tensor class satisfies

$$\mathbb{E}_0|C| \ge \Delta \left[1 - 2\alpha - \frac{1}{2\sqrt{H_d(\Delta)}}\right]_+.$$

Proof. The Gaussian likelihood identity gives overlap 1 for distinct orthogonal alternatives and $1 + v_i$ for equal alternatives. Subtracting 1 leaves $\sum \pi_i^2 v_i$. Cauchy–Schwarz bounds this below by $(\sum v_i^{-1})^{-1}$ and the

displayed weights attain equality. Apply TV $\le \sqrt{\chi^2}/2$ and the common-target interval lemma. This uses the full observation; there is no diagonal-data restriction.

`weighted_axis_lower` enumerates d and uses scalar root finding for the largest $\Delta \le M w_d$ with $\chi^2 \le 1/4$. Every reported lower bound uses the actual recomputed divergence and a feasible prior. This optimizes the prior at each chosen target within the axis family. It is not an optimization over all tensor alternatives. For equal weights it recovers the one-spike logarithmic obstruction. It cannot recover the cubic harmonic obstruction on its own and is not claimed sharp for all weights.

A1. Unseparated rate adaptation across positive fixed caps

Fix an outer positive Tucker or TT cap vector $\bar{r}$. Let $\mathcal{R}$ contain any collection of positive cap vectors componentwise below it. For equal or flat orthogonal loadings, the *same* full-data outer-class interval satisfies

$$\sup_{A \in \Theta(M,r)} \mathbb{E}_A |C_{\bar{r}}| \le C_{K,\bar{r},\alpha}\, \mathfrak{r}_K(m, M, \tau), \qquad r \in \mathcal{R}.$$

Every smaller class contains the rank-one ball, whose lower bound is $c_{K,\alpha}\mathfrak{r}_K$. Therefore the interval is simultaneously minimax in rate over every such class, uniformly in m, M, τ, without knowing the actual ranks or imposing singular-value separation. The analogous MSE conclusion holds for the outer-class ideal residual fit.

Proof. Restrict the existing outer supremum upper bound to each subset and apply the same rank-one lower bound inside each smaller class. No selection step or new probability argument is required.

This observation resolves the fixed-positive-cap, rate-only statistical adaptation question. It does not give optimal rank-dependent constants, growing-cap rates, computation, exact-rank selection, or pointwise oracle lengths. Rank zero must be excluded: its known-model oracle interval has length zero, whereas outer-class honesty forces positive length at zero. The preceding ledgers used the word adaptation broadly; these distinct objectives must be separated.

A2. A computable projection certificate without separation

Let $Z^{(1)} = A + E^{(1)}$ and $Z^{(2)} = A + E^{(2)}$ have independent iid Gaussian errors of known scales τ_1, τ_2. Only $\|A\|_F \le M$ is assumed here; supplied rank caps are unnecessary. Let $0 < \delta < \alpha$. Use the simultaneous unfolding operator envelopes from the previous stage:

$$\eta_k = 2\tau_1 \sqrt{2\{(p_k + \prod_{l \ne k} p_l)\log 9 + \log(2K/\delta)\}}.$$

On one event of probability at least $1 - \delta$ all $\|E^{(1)}_{(k)}\|_{op} \le \eta_k$. For any pilot-chosen ranks r_k, let Q_k project onto the first r_k left singular vectors of $Z^{(1)}_{(k)}$. Set $d_k = \sigma_{r_k+1}(Z^{(1)}_{(k)}) + \eta_k$ if $r_k < p_k$, and $d_k = 0$ if $Q_k = I$. Singular values beyond the matrix rank are zero. For $P = \bigotimes_k Q_k$, decompose

$$R = (I - P)L = \sum_{k=1}^{K} R_k, \quad R_k = \left[\bigotimes_{l<k} Q_l\right] \otimes (I - Q_k) \otimes \left[\bigotimes_{l>k} I\right] L.$$

Then the observable bound

$$B = \min\left\{ M\|R\|_F,\ \sum_k d_k \|(R_k)_{(k)}\|_* \right\}$$

satisfies $|\langle R, A\rangle| \le B$ on that event. The interval centered at $\langle PL, Z^{(2)}\rangle$, with radius

$$B + z_{1-(\alpha-\delta)/2}\tau_2 \|PL\|_F,$$

is honest over the whole norm ball. Any finite set of pilot-selected projection choices can be compared by their radii on this same event; choose the smallest before using validation data. Include $P = I$ for the direct interval and $P = 0$ for the structural interval. Intersect with $[-M\|L\|_F, M\|L\|_F]$ and use $\{0\}$ for an empty intersection.

Proof. Residual SVD optimality and the triangle inequality give $\|(I - Q_k)A_{(k)}\|_{op} \le d_k$ on the common event, even for selected r_k. The definition of R_k gives $(I - Q_k)(R_k)_{(k)} = (R_k)_{(k)}$. Operator/nuclear norm duality bounds its contraction with A by $d_k\|(R_k)_{(k)}\|_*$. Sum and combine with the norm-only bound. Conditional on the pilot, validation noise in the selected loading is Gaussian with variance $\tau_2^2\|PL\|_F^2$. No multiplicity correction over the candidate ranks is required: all bias certificates hold on one uniform pilot event, and the choice precedes validation. The union bound gives total error at most α. Unfolding SVDs, tensor projections and nuclear norms are polynomial in the explicit input size.

This algorithm never interprets an undetected component as identically zero. It does not certify the actual rank. Its bias term can be large, and no weak-signal or pointwise minimax optimality is claimed. Honesty on the full norm ball cannot yield the low-rank weak-signal minimax gain: the earlier unrestricted-class lower bound still applies. Independent artificial Gaussian splitting of one observation is valid with $\tau_1 = \tau_2 = \sqrt{2}\tau$; using both halves at scale τ is invalid.

Why the first-order repair does not solve the original problem

For the A2 construction, if $\min_k \eta_k \ge q\tau_2$, where $q = z_{1-(\alpha-\delta)/2}$, no projected candidate improves the smaller of the untruncated direct and norm-only radii. In fact, nonzero R_k have $d_k \ge \eta_k$, so triangle inequalities and nuclear/Frobenius norm comparison give $B \ge \min(M, \min_k \eta_k)\|R\|_F$. Hence its radius is at least $\min(M, q\tau_2)(\|PL\|_F + \|R\|_F)$, which is at least $\min(M, q\tau_2)\|L\|_F$. The explicit envelopes meet this condition in the equal-noise experiments below. This is a proved limitation of this particular repair, not a universal computational lower bound.

A3. The adaptation obstruction at an arbitrary orientation

Let a base tensor A_0 have norm $a < M$, and leave at least one unused unit of every supplied Tucker or TT rank cap. Choose mode subspaces W_k such that A_0 is orthogonal to every tensor in $\mathcal{W} = \bigotimes_k W_k$. One sufficient choice is to take at least one W_k orthogonal to the column space of $A_{0,(k)}$. Put $M_0 = \sqrt{M^2 - a^2}$ and $L_{\mathcal{W}} = P_{\mathcal{W}}L$. Then

$$\mathbb{E}_{A_0}|C| \ge c_\alpha \min(M_0, \tau)\rho(L_{\mathcal{W}})$$

for every interval honest on the outer class. If the restriction of the loading to an m-dimensional orthogonal diagonal block in $\mathcal{W}$ equals b times a flat unit loading, the stronger bound is

$$\mathbb{E}_{A_0}|C| \ge c_{K,\alpha}|b|\,\mathfrak{r}_K(m, M_0, \tau).$$

Here ρ is taken within the indicated product subspace. The zero overlap case gives a zero lower bound and says nothing further.

Proof. Add rank-one perturbations $B \in \mathcal{W}$ with norm at most M_0. Rank subadditivity keeps $A_0 + B$ in every outer unfolding cap; orthogonality gives $\|A_0 + B\|_F^2 = a^2 + \|B\|_F^2 \le M^2$. Translate the Gaussian experiment by A_0. The target difference is $\langle L_{\mathcal{W}}, B\rangle$. Apply the two-point construction of L1 or the existing flat-loading mixtures, all of which have rank one. Unused observed coordinates contain identical independent Gaussian noise under both hypotheses and do not change the likelihood calculation. This is a rotation-invariant extension of the preceding block-supported shifted bound, not a complete characterization at every mean.

A4. A cap-valid tangent interval on the original unseparated class

There is a more useful repair when mode caps $\bar{r}_k \le p_k$ are supplied. Use the same independent Gaussian pilot and validation data and envelopes as A2. Let Q_k project onto the leading $\bar{r}_k$ pilot left singular vectors. Define

$$e_k = \begin{cases} 0, & \bar{r}_k = p_k, \\ \min\{1, 2\eta_k/(\sigma_{\bar{r}_k}(Z^{(1)}_{(k)}) - \eta_k)\}, & \bar{r}_k < p_k,\ \sigma_{\bar{r}_k}(Z^{(1)}_{(k)}) > \eta_k, \\ 1, & \text{otherwise.} \end{cases}$$

With the first-order product-space projector from the previous stage,

$$T = \bigotimes_k Q_k + \sum_k (I - Q_k) \otimes \bigotimes_{l \neq k} Q_l,$$

put $R = (I - T)L$ and

$$B_T = M\|R\|_F \min\{1, (\sum_{k<l} e_k^2 e_l^2)^{1/2}\}.$$

The interval centered at $\langle TL, Z^{(2)} \rangle$, with radius $B_T + q\tau_2\|TL\|_F$, is honest on **every** tensor with these supplied mode caps and norm bound, including zero, arbitrarily weak signals, and actual ranks strictly below the caps. No singular-value separation is assumed. Choose the smallest pilot-measurable radius among this candidate, the direct interval, and the structural norm interval, as before.

Proof. Let V_k be the actual left singular subspace of $A_{(k)}$. On the simultaneous noise event we claim $\|(I - Q_k)P_{V_k}\|_{op} \leq e_k$. This is immediate for $e_k = 1$ or $Q_k = I$. Otherwise $\sigma_{\bar{r}_k}(Z^{(1)}_{(k)}) > \eta_k$ forces the true mode rank to equal the supplied cap: a smaller true rank would give that observed singular value at most η_k by Weyl's inequality. The pilot tail after the cap is at most η_k, so $\|(I - Q_k)A_{(k)}\|_{op} \leq 2\eta_k$. Multiplying by the inverse nonzero singular values and using $\sigma_{\bar{r}_k}(A_{(k)}) \geq \sigma_{\bar{r}_k}(Z^{(1)}_{(k)}) - \eta_k$ proves the claim.

Decompose A into the mutually orthogonal mode-projector blocks indexed by which modes fall in $I - Q_k$. The complement of T consists of blocks with at least two such modes. Counting each block once is bounded by counting it for every contained pair. Consequently

$$\|(I - T)A\|_F^2 \leq \sum_{k<l} \|(I - Q_k)(I - Q_l)A\|_F^2 \leq M^2 \sum_{k<l} e_k^2 e_l^2.$$

The last inequality restricts each operator to its true mode subspace and uses that operators on distinct modes commute. Combine with the trivial bound $\|(I - T)A\|_F \leq M$ and Cauchy–Schwarz against R. Conditional Gaussian validation and the common pilot event now prove coverage and valid radius selection exactly as in A2.

This algorithm is polynomial and honest for the original weak-signal class. Strong signals can make the second-order bias correction small, even though separation is not a coverage assumption. It is **not** a polynomial weak-signal minimax solution or a method of estimating actual rank. At smaller actual ranks, the unresolved cap singular values trigger conservative error bounds. For a supplied TT cap vector, valid mode caps are $\min(p_k, \bar{r}_{k-1}\bar{r}_k)$, with boundary TT ranks one; the interval does not estimate TT bond ranks. Artificial splitting must use its increased noise scales.

For the original single observation, `single_observation_interval` creates $Z \pm \tau G'$ with independent artificial G', uses scales $\sqrt{2}\tau$ and error $\alpha/2$ for the cap-valid construction, and intersects with the direct interval based on Z at error $\alpha/2$. The union bound proves honesty at error α despite the dependence between these two intervals. The direct component caps the reported length at $2z_{1-\alpha/4}\tau\|L\|_F$. The split is randomization within the original experiment, not an extra observed replicate.

Remaining problems

1. Polynomial minimax intervals for diffuse weak-signal classes at general dimensions: open. The new optimizer establishes a working global benchmark only for binary rank-one tensors; it does not remove the computational obstruction.

2. Sharp larger-budget rates for every loading, including every unequal weight sequence: open. L1 resolves all loadings only for $M \leq \tau$; L2 strengthens the usable lower-bound machinery.
3. General rank-dependent and pointwise adaptation: open, with explicit impossibility results for some oracle demands. Rate adaptation over fixed positive nested caps is already attained statistically (A1).
4. General Tucker/TT global optimizers, validated directed-rounding arithmetic, and high-dimensional runtime scaling: not implemented.